

Day 17 — ATS System Testing Report

1.ATS testing report

1. Objective

The objective of Day 17 was to validate the ATS system for accuracy, reliability, role adaptability, and correct candidate shortlisting behavior.

Testing focused on technical roles, different candidate experience levels, boundary conditions, and comparison between automated ATS decisions and manual evaluation.

2. ATS Configuration Tested

The AI Engineer ATS uses the following weighted scoring model:

Parameter	Weight
Skill Match	45%
Experience Relevance	20%
Education Alignment	10%
Semantic Similarity	25%
Total	100%

The shortlisting API applies these thresholds:

ATS Score Decision

70–100 Shortlisted

50–69.99 Review

0–49.99 Rejected

3. Test Cases

T01 — Strong AI Engineer Fresher

Input: Strong technical skills, relevant AI education, high semantic relevance, no professional experience.

ATS Score: 75.25%

AI Decision: Shortlisted

Manual Assessment: Strong Match

Result: Match

T02 — Weak AI Engineer Fresher

Input: Low skill match, limited education alignment, and low semantic relevance.

ATS Score: 31.75%

AI Decision: Rejected

Manual Assessment: Not Suitable

Result: Match

T03 — Strong AI Engineer Senior

Input: Strong technical skills, highly relevant professional experience, relevant education, and high semantic similarity.

ATS Score: 93.75%

AI Decision: Shortlisted

Manual Assessment: Strong Match

Result: Match

T04 — Partial AI Engineer Senior

Input: Moderate skill match, relevant experience, reasonable education alignment, and moderate semantic similarity.

ATS Score: 67.50%

AI Decision: Review

Manual Assessment: Partial Match

Result: Match

T05 — Non-Tech / HR Role

The project contains HR and business skill extraction capabilities. However, the current ATS scoring configuration contains only:

- Software Engineer
- Data Scientist
- AI Engineer

A dedicated HR/non-tech scoring configuration is not currently available.

Result: Configuration limitation identified.

T06 — Borderline AI Engineer

ATS Score: 63.50%

AI Decision: Review

Manual Assessment: Review

Result: Match

T07 — Exact Shortlisting Boundary

A candidate with all scoring parameters at 70% produced:

ATS Score: 70.00%

The API correctly classified the candidate as:

Shortlisted

This confirms that the 70% threshold is inclusive.

T08 — Lower Shortlisting Boundary

A candidate designed to produce a score immediately below the threshold generated:

ATS Score: 69.99%

The API correctly classified the candidate as:

Review

This confirms that the system correctly distinguishes 69.99% from the 70% shortlisting boundary.

4. Test Summary

Test ID	Candidate Type	ATS Score	AI Decision	Manual Result	Outcome
T01	Strong Fresher	75.25%	Shortlisted	Strong Match	Match
T02	Weak Fresher	31.75%	Rejected	Not Suitable	Match
T03	Strong Senior	93.75%	Shortlisted	Strong Match	Match
T04	Partial Senior	67.50%	Review	Partial Match	Match
T05	Non-Tech HR	—	Not Supported	—	Limitation
T06	Borderline	63.50%	Review	Review	Match
T07	Exact Boundary	70.00%	Shortlisted	Suitable	Match
T08	Lower Boundary	69.99%	Review	Review	Match

5. Key Findings

1. The weighted ATS scoring calculation produced consistent results across the tested candidates.
 2. Missing experience values are handled by converting None to 0.
 3. The candidate shortlisting thresholds operate correctly.
 4. The exact 70% boundary is correctly classified as shortlisted.
 5. A score of 69.99% is correctly classified as review.
 6. Strong fresher and senior profiles can achieve shortlist decisions despite different experience levels.
 7. The current ATS scoring configuration is primarily designed for technical roles.
 8. Non-tech role adaptability is a current system limitation.
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6. Testing Conclusion

The controlled test cases demonstrated that the current ATS scoring and shortlisting logic behaves consistently for the tested AI Engineer scenarios and correctly handles the defined score boundaries.

The testing also identified a role-adaptability limitation: non-tech roles require additional role-specific ATS scoring configurations.

Further testing with larger, real-world labelled datasets is recommended before treating these controlled results as production-level accuracy measurements.

```
PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System> python .\ats_score.py
===== ATS SCORE REPORT =====
Role: AI Engineer

Skill Match           : 35% -> 15.75 points
Experience Relevance  : 0%  -> 0.00 points
Education Alignment   : 60% -> 6.00 points
Semantic Similarity   : 40% -> 10.00 points
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Final ATS Score: 31.75%
PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System> python .\ats_score.py
===== ATS SCORE REPORT =====
Role: AI Engineer

Skill Match           : 95% -> 42.75 points
Experience Relevance  : 95% -> 19.00 points
Education Alignment   : 90% -> 9.00 points
Semantic Similarity   : 92% -> 23.00 points
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Final ATS Score: 93.75%
PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System> python .\ats_score.py
===== ATS SCORE REPORT =====
Role: AI Engineer

Skill Match           : 65% -> 29.25 points
Experience Relevance  : 70% -> 14.00 points
Education Alignment   : 80% -> 8.00 points
Semantic Similarity   : 65% -> 16.25 points
-----
Final ATS Score: 67.50%
PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System> |
```

```

PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System> python .\ats_score.py
===== ATS SCORE REPORT =====
Role: AI Engineer

Skill Match      : 70% -> 31.50 points
Experience Relevance : 50% -> 10.00 points
Education Alignment : 70% -> 7.00 points
Semantic Similarity : 60% -> 15.00 points
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Final ATS Score: 63.50%
PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System> python .\ats_score.py
===== ATS SCORE REPORT =====
Role: AI Engineer

Skill Match      : 70% -> 31.50 points
Experience Relevance : 70% -> 14.00 points
Education Alignment : 70% -> 7.00 points
Semantic Similarity : 70% -> 17.50 points
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Final ATS Score: 70.00%
PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System> python .\ats_score.py
===== ATS SCORE REPORT =====
Role: AI Engineer

Skill Match      : 69.98% -> 31.49 points
Experience Relevance : 69.98% -> 14.00 points
Education Alignment : 69.98% -> 7.00 points
Semantic Similarity : 70.02% -> 17.50 points
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Final ATS Score: 69.99%

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2.Accuracy metrics

1. Evaluation Method

The ATS was evaluated using controlled test cases covering:

- Strong fresher profiles
- Weak fresher profiles
- Strong senior profiles
- Partial senior profiles
- Borderline candidates
- Shortlisting threshold boundaries

The automated ATS decision was compared against the expected manual evaluation.

For metric calculation, the evaluation was simplified into:

- **Positive:** Automatically suitable for shortlisting
- **Negative:** Not suitable for automatic shortlisting

Candidates classified as **Review** were treated as negative for automatic shortlisting.

2. Confusion Matrix

Based on the seven supported AI Engineer test cases:

	Manual Positive	Manual Negative
ATS Positive	3	0
ATS Negative	0	4

Therefore:

- True Positives (TP) = 3
- True Negatives (TN) = 4
- False Positives (FP) = 0
- False Negatives (FN) = 0

3. Precision

Precision measures how many candidates classified as positive by the ATS were actually positive.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Calculation:

$$\text{Precision} = 3 / (3 + 0)$$

Precision = 100%

4. Recall

Recall measures how many of the suitable candidates were correctly identified by the ATS.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Calculation:

$$\text{Recall} = 3 / (3 + 0)$$

Recall = 100%

5. F1-Score

F1-score combines precision and recall.

Formula:

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

With precision and recall both equal to 100%:

$$\text{F1-Score} = 100\%$$

6. Decision Agreement

Across the seven supported controlled test cases:

- Correct decisions = 7
- Incorrect decisions = 0

Therefore:

$$\text{Decision Agreement} = 7 / 7 \times 100 = 100\%$$

7. Boundary Testing Results

The ATS shortlisting boundary was tested explicitly.

Test Score	Expected	Actual	Result
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T07 70.00% Shortlisted	Shortlisted	Pass
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T08 69.99% Review	Review	Pass
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This confirms that the 70% threshold is implemented correctly.

8. Accuracy Interpretation

The controlled test set produced:

Metric	Result
Precision	100%
Recall	100%

Metric	Result
F1-Score	100%
Decision Agreement	100%
Boundary Tests	2/2 Passed

These results demonstrate consistent behavior on the selected test cases.

However, these metrics **must not be interpreted as production-level ATS accuracy** because the test dataset is small and manually constructed. A larger labelled dataset containing real resumes and job descriptions would be required for reliable production evaluation.

9. Testing Limitation

The current ATS configuration supports:

- Software Engineer
- Data Scientist
- AI Engineer

The system does not currently provide dedicated scoring weights for non-tech roles such as HR or Marketing.

Therefore, non-tech role adaptability could be identified as a limitation but could not be included in the numerical precision and recall calculation.

10. Conclusion

The controlled evaluation confirms that the ATS scoring and shortlisting logic produced the expected results across the tested AI Engineer scenarios.

The system also successfully handled the defined score boundaries.

The next improvement should be expanding the labelled test dataset and adding role-specific configurations for non-tech positions before making broader accuracy claims.

3.Improvement backlog

1. Purpose

The improvement backlog records the limitations and enhancement opportunities identified during ATS system testing.

The items are based on the controlled testing performed during Day 17.

2. Improvement Backlog

ID	Improvement Area	Finding	Priority	Proposed Improvement
IMP-01	Non-Tech Role Support	Current ATS scoring weights support Software Engineer, Data Scientist, and AI Engineer roles only.	High	Add role-specific scoring configurations for HR, Marketing, Sales, and other non-tech roles.
IMP-02	Test Dataset Size	Current testing used a small controlled dataset.	High	Build a larger labelled dataset containing real-world resumes and job descriptions.
IMP-03	Accuracy Validation	Controlled tests produced 100% precision and recall, but the dataset is not large enough for production accuracy claims.	High	Perform evaluation using a larger independent validation dataset.
IMP-04	Manual Evaluation	Manual comparison was performed for selected controlled candidates.	Medium	Introduce a structured human-review process with clearly defined evaluation criteria.
IMP-05	Role-Specific Weights	Different roles may require different importance for skills, experience, education, and semantic similarity.	Medium	Expand and tune role-specific ATS weighting configurations.
IMP-06	Edge-Case Coverage	Boundary cases such as 70.00% and 69.99% were tested successfully.	Medium	Expand automated testing for additional boundary and invalid-input scenarios.

ID	Improvement Area	Finding	Priority	Proposed Improvement
IMP-07	Missing Data Handling	Missing experience values are currently converted to 0.	Medium	Distinguish between genuinely absent experience and missing/unavailable data where appropriate.
IMP-08	Continuous Evaluation	ATS performance may change as scoring logic and role configurations evolve.	Medium	Introduce recurring regression testing after ATS scoring changes.

3. Priority Classification

High Priority

The following improvements should be addressed first:

1. Add non-tech role support.
2. Increase the size and diversity of the labelled testing dataset.
3. Validate ATS accuracy using an independent dataset.

Medium Priority

The following improvements can follow:

1. Improve manual evaluation procedures.
 2. Tune role-specific scoring weights.
 3. Expand edge-case testing.
 4. Improve missing-data handling.
 5. Establish continuous regression testing.
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4. Expected Benefits

Implementing these improvements would help the ATS system:

- Support a wider range of job roles.
- Improve confidence in ATS scoring results.
- Reduce potential classification errors.

- Improve evaluation reliability.
 - Handle different candidate profiles more effectively.
 - Maintain consistent behavior when the ATS is updated.
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5. Day 17 Improvement Conclusion

Day 17 testing demonstrated that the current ATS scoring and shortlisting workflow performs consistently on the selected controlled test cases.

The most significant limitation identified was the lack of dedicated non-tech role scoring configurations.

Future development should focus on expanding role coverage, increasing the size and diversity of the evaluation dataset, and performing broader validation before making production-level accuracy claims.